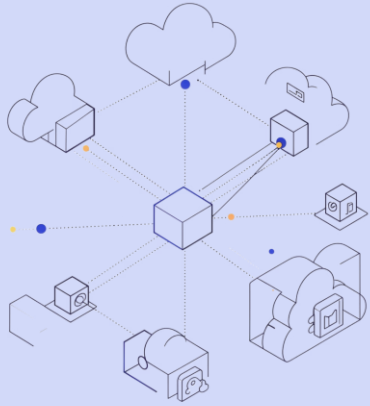




CloudCostLens

Intelligent AWS Architecture Advisor - recommending cost-optimal architectures, automating provisioning, and eliminating runaway cloud bills.

Project Overview

What It Is

CloudCostLens is a platform that analyzes your traffic patterns and budget constraints, then recommends and automatically provisions the most cost-optimal AWS architecture for your workload.

The Problem It Solves

- **Runaway cloud bills** — resources are auto-destroyed after 2 hours, enforcing hard cost boundaries
- **Manual provisioning complexity** — eliminates tedious, error-prone AWS console workflows





DEVSECOPS OPTIMIZATION

Automated Cloud Infrastructure

Get intelligent AWS recommendations, then provision real infrastructure with a single click — powered by Terraform modules and production-grade DevOps practices.

[Start Analysis](#)[Technical Overview](#)

Deployment History

Database record of all infrastructure lifecycle events.

14

TOTAL

0

COMPLETED

4

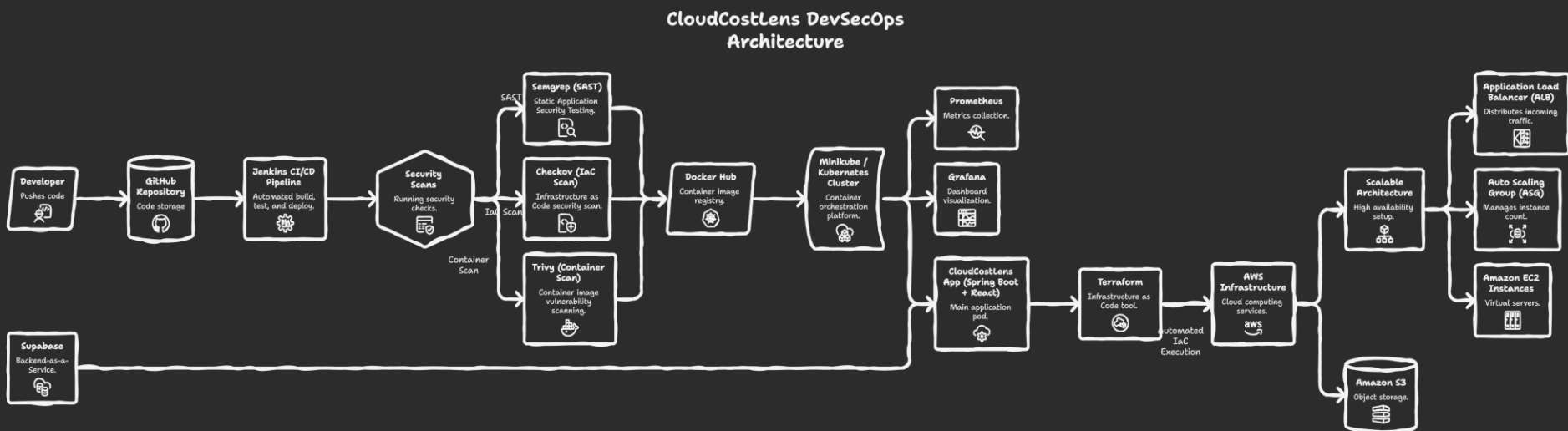
FAILED

3

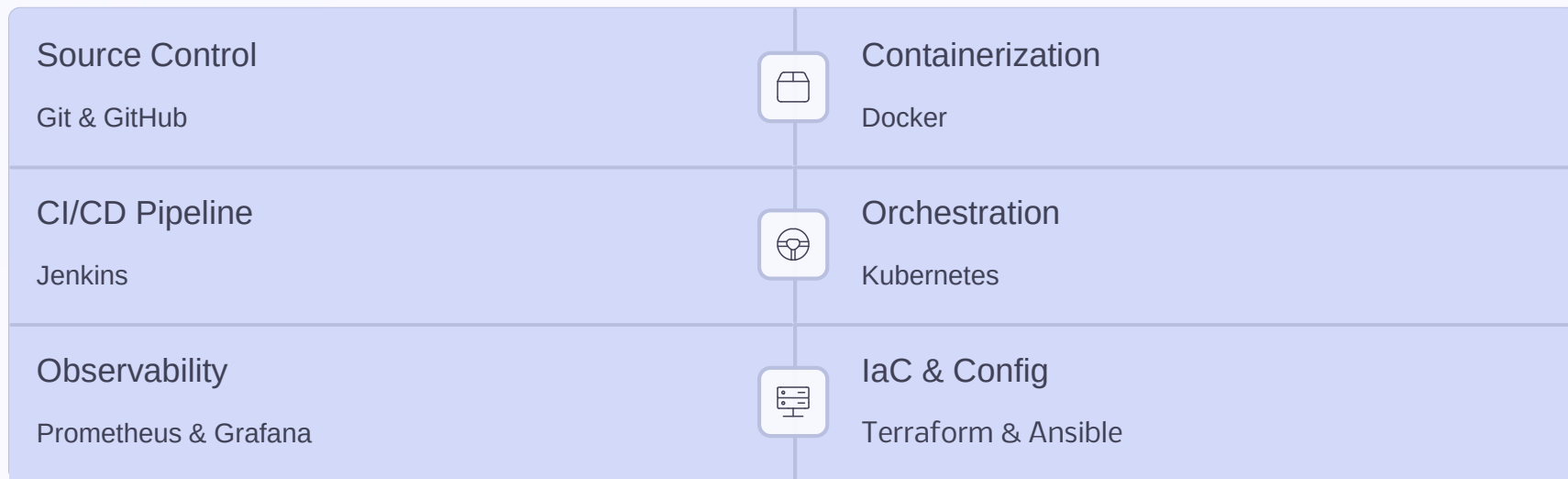
DESTROYED

DEPLOYMENT	MODULE	REGION	STATUS	COST	CREATED	DURATION	ACTIONS
c8d68551 full-stack	[W] web_app	us-east-1	PLANNING	—	11 Apr 2026, 08:41 pm	—	View 🔍
812c5dfe full-stack	[W] web_app	us-east-1	PLANNING	—	11 Apr 2026, 08:41 pm	—	View 🔍
98775e11 backend-api	[W] web_app	us-east-1	PLANNING	—	11 Apr 2026, 07:00 pm	—	View 🔍
97c88976 backend-api	[W] web_app	us-east-1	DESTROYED	\$3.50/mo (≈ ₹294)	11 Apr 2026, 06:37 pm	16m 43s	View 🔍
b5212cba backend-api	[W] web_app	us-east-1	DESTROYED	\$3.50/mo (≈ ₹294)	10 Apr 2026, 03:47 pm	124m 53s	View 🔍

Architecture



Full DevSecOps Stack



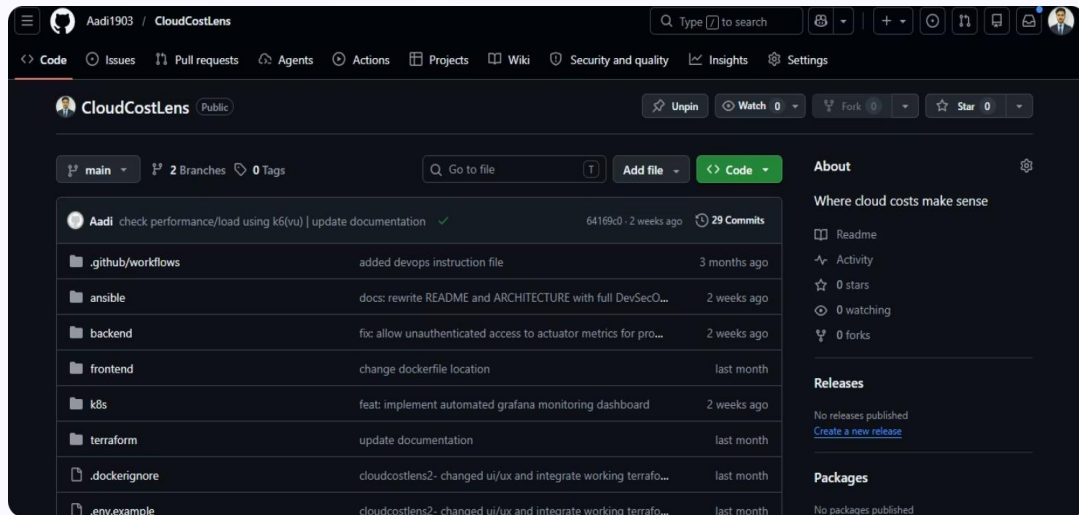
Git & GitHub

What It Is

Industry-standard version control and source code management for collaborative software development.

Role in CloudCostLens

- Enforces a strict **main vs. dev branching strategy** to protect production code
- Every commit **automatically triggers** downstream Jenkins CI/CD pipelines
- Provides full audit trail and rollback capability across the entire codebase



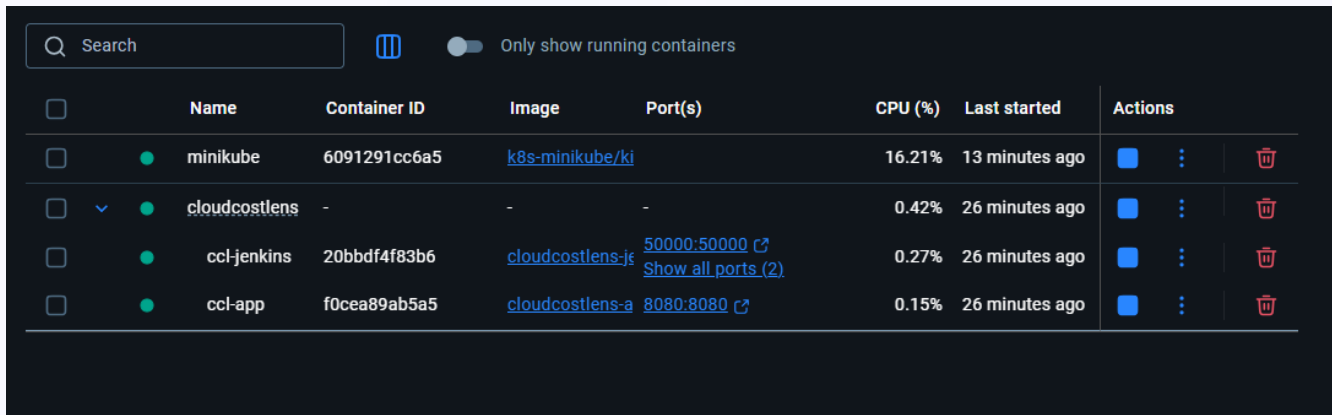
Docker

What It Is

A leading containerization platform that packages applications and their dependencies into portable, isolated units.

Role in CloudCostLens

- Packages the **Spring Boot backend** and **React frontend** into lightweight container images
- Guarantees consistent behavior across dev, staging, and production environments
- Serves as the foundational artifact consumed by the Kubernetes orchestration layer



The screenshot shows the Docker Desktop interface with a dark theme. At the top, there is a search bar and a toggle switch labeled 'Only show running containers'. Below this is a table of running containers. The table has columns for Name, Container ID, Image, Port(s), CPU (%), Last started, and Actions. There are four containers listed: minikube, cloudcostlens, ccl-jenkins, and ccl-app. Each container row has a checkbox, a status indicator (green dot), and a trash icon in the Actions column.

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	minikube	6091291cc6a5	k8s-minikube/ki		16.21%	13 minutes ago	☐ ⋮
☐	cloudcostlens	-	-	-	0.42%	26 minutes ago	☐ ⋮
☐	ccl-jenkins	20bbdf4f83b6	cloudcostlens-jenkins	50000:50000 ↗ Show all ports (2)	0.27%	26 minutes ago	☐ ⋮
☐	ccl-app	f0cea89ab5a5	cloudcostlens-app	8080:8080 ↗	0.15%	26 minutes ago	☐ ⋮

Insert your Docker screenshot here.

Jenkins

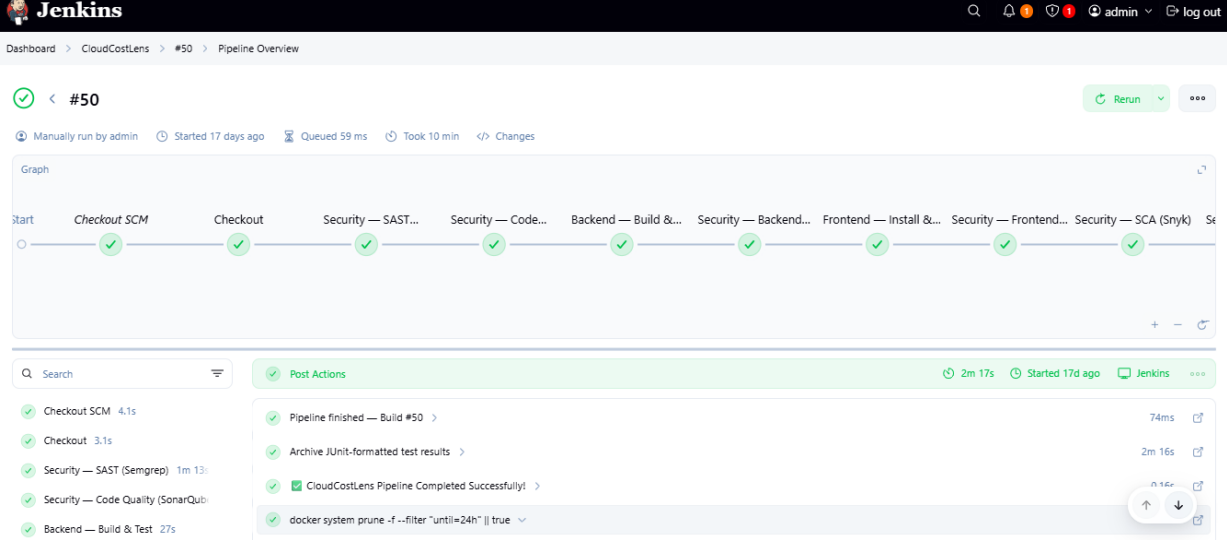
What It Is

An open-source automation server that orchestrates continuous integration and continuous delivery pipelines.

Role in CloudCostLens

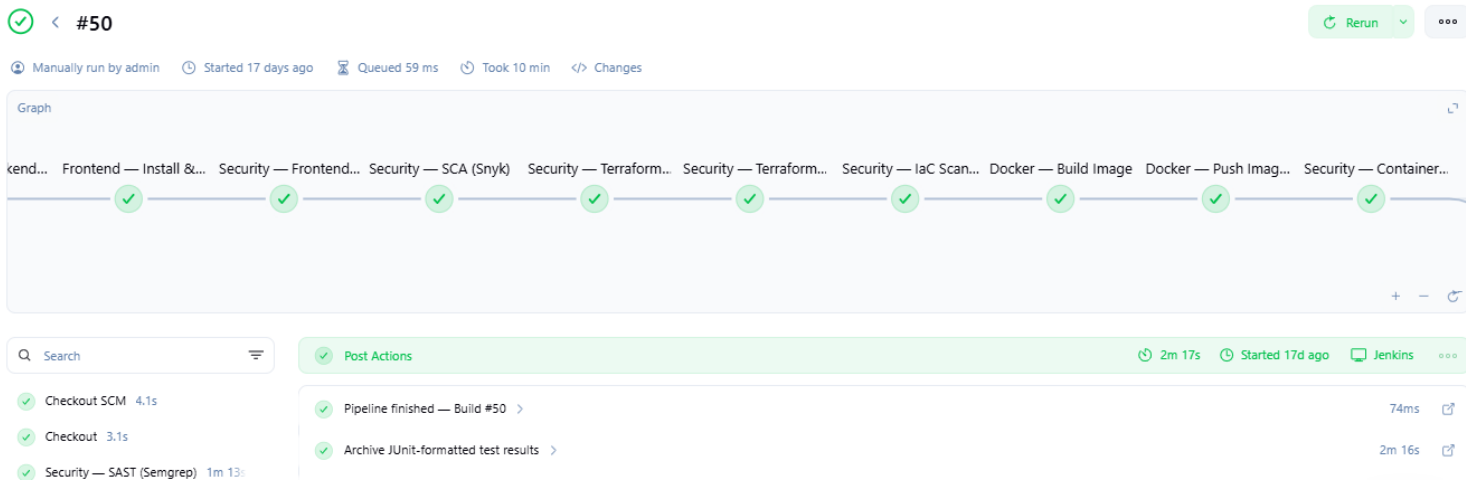
- Drives the full **DevSecOps pipeline** — from code compilation to deployment
- Runs **SAST and SCA security scans** automatically on every build
- Triggers containerized deployments upon successful pipeline completion

Screenshot Placeholder



There are **17 stages** defined in Jenkins pipeline.

It is a highly secure DevSecOps pipeline that integrates security scans at almost every step of the lifecycle.



Kubernetes (K8s)

What It Is

A production-grade container orchestration system for automating deployment, scaling, and management of containerized workloads.

Role in CloudCostLens

- Provides **self-healing and auto-scaling** for all Docker containers
- Manages service discovery and internal networking via **Kubernetes Services**
- Securely injects app configuration using **ConfigMaps and Secrets**

```
● PS D:\Documents\Mini Project\cloudcostlens> kubectl apply -f k8s/
configmap/ccl-config unchanged
deployment.apps/ccl-app-deployment unchanged
configmap/grafana-datasource-config unchanged
configmap/grafana-dashboard-provider unchanged
configmap/grafana-dashboard-json unchanged
persistentvolumeclaim/grafana-data-pvc unchanged
deployment.apps/grafana-deployment unchanged
service/grafana-service unchanged
namespace/cloudcostlens unchanged
configmap/prometheus-config unchanged
persistentvolumeclaim/prometheus-data-pvc unchanged
deployment.apps/prometheus-deployment unchanged
service/prometheus-service unchanged
persistentvolumeclaim/terraform-data-pvc unchanged
secret/ccl-secret configured
service/ccl-app-service unchanged
● PS D:\Documents\Mini Project\cloudcostlens> kubectl get pods -n cloudcostlens
NAME                                READY   STATUS    RESTARTS   AGE
ccl-app-deployment-54d46d746f-lzc8d  1/1     Running   4 (4d ago)  17d
grafana-deployment-68bc9d9fc6-2qm46  1/1     Running   3 (4d ago)  16d
prometheus-deployment-6d487d4d85-zk69v 1/1     Running   4 (4d ago)  17d
```

Prometheus & Grafana

What It Is

An industry-standard monitoring and observability stack for collecting, storing, and visualizing real-time performance metrics.

Role in CloudCostLens

- **Prometheus** scrapes live metrics — CPU usage, database connections, JVM health — from the Spring Boot app
- **Grafana** renders those metrics on interactive, real-time dashboards
- Enables proactive alerting before performance degradation impacts users

Screenshot Placeholder

Insert your Grafana dashboard screenshot here.



Prometheus Target Health monitor showing a healthy, active connection (UP state) to backend application.

This ensures continuous, automated collection of application metrics to feed real-time Grafana dashboards.

 Prometheus

Query

Alerts

Status > Target health

Select scrape pool

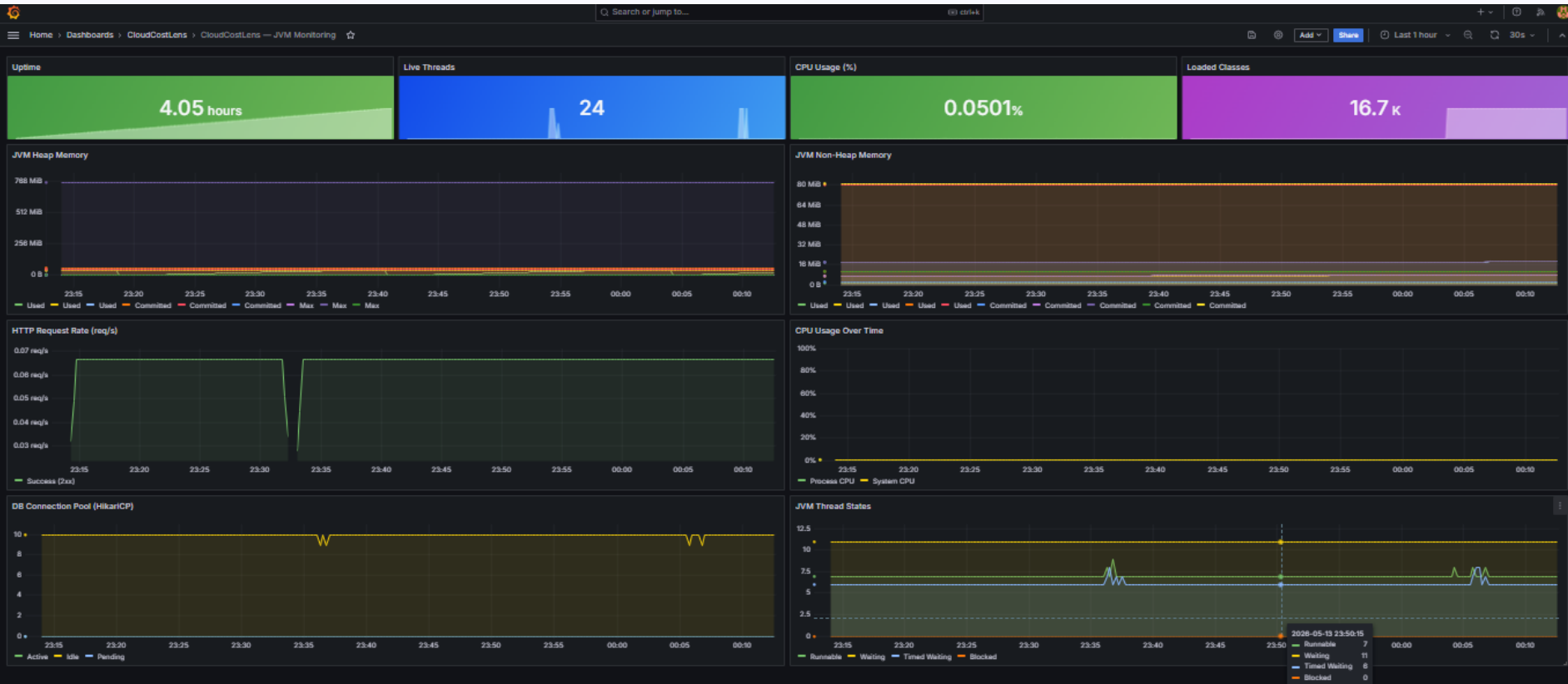
Filter by target health

Filter by endpoint or labels

cloudcostlens-backend1 / 1 up

Endpoint	Labels	Last scrape	State
http://ccl-app-service.cloudcostlens.svc.cluster.local:8080/actuator/prometheus	instance="ccl-app-service.cloudcostlens.svc.cluster.local:8080" job="cloudcostlens-backend"	5.765s ago 39ms	UP

Grafana Dashboard



Terraform & Ansible

What They Are

Terraform is an IaC tool for provisioning cloud infrastructure declaratively.

Ansible is a configuration management tool for automating server setup.

Role in CloudCostLens

- **Terraform** creates all required AWS resources on demand
- **Ansible** automates security hardening and configuration of host servers
- Together they enable fully **repeatable, auditable infrastructure** deployments

Screenshot Placeholder

Insert your Terraform/Ansible screenshot here.

THAT'S ALL